

CEN 204 Midterm Exam Solutions

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Sat

① $1 \leq q < 2$

$$x = 2^5 + 2^{-10} + 2^{-19}$$

$$x = 2^5(1 + 2^{-15} + 2^{-24})$$

$$x = 1.000000\dots 1\dots 1 \cdot 2^5$$

$a_{15} \quad a_{24}$

— it is chopping

$$x_- = (1.0\dots 1\dots 1)_2 \cdot 2^{-5} = 2^{-5}(1 + 2^{-15})$$

a_{15}

$$x_+ = (1.00\dots 1\dots 1)_2 \cdot 2^5 = 2^5(1 + 2^{-15} + 2^{-23})$$

$a_{15} \quad a_{24}$

② $x = 0.1643676895$

$(x+y)$ $y = 0.1643671734$

$$\left| \frac{(x+y) - [8(x) + 8(y)]}{x+y} \right|$$

$$x+y = 1.28734863 \rightarrow \text{real}$$

$$x+y = 1.287349 \rightarrow \text{measured}$$

$$\text{Absolute} = |\text{real} - \text{measured}| = 0.00000037$$

$$\text{Relative} = \left| \frac{\text{Absolute}}{\text{Real}} \right| \times 100 = \frac{0.00000037}{1.28734863} \times 100$$

% 0.000028

$(x-y)$ $\left| \frac{(x-y) - [8'(x) - 8'(y)]}{x-y} \right|$

$$= \left| \frac{(0.0000005161 - 0.0000005)}{0.0000005161} \right| \times 100$$

= % 3.11

3

$$f(x) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = 1 \quad x_1 = -1$$

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

$$x_2 = -1 - 0 \cdot \left(\frac{-1-1}{0-(-8)} \right) = -1 + 0 = \boxed{-1}$$

4) $f(x) = x^3 + 2x^2 - 5x - 6$

$$x_0 = 1$$

$$f'(x) = 3x^2 + 4x - 5$$

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{1+2-5-6}{3+4-5} = 1 - \frac{-8}{2} = \boxed{5}$$

$$x_2 = 5 - \frac{5^3 + 2 \cdot 5^2 - 5 \cdot 5 - 6}{3 \cdot 5^2 + 4 \cdot 5 - 5}$$

$$x_2 = 5 - \frac{144}{90} = \boxed{3,4}$$